# Kun-Ying Lee

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Robot Learning | Humanoid Locomotion | Motion Generation

## Education

**Tsinghua University** 2025–2029 (expected)
**Undergraduate, Yao Class** GPA: 3.8/4.0
Computer Science (Artificial Intelligence Track)
Institute for Interdisciplinary Information Sciences (IIIS)

## Selected Publications

**OMG: Omni-Modal Motion Generation for Generalist Humanoid Control**
Siqiao Huang*, **Kun-Ying Lee***, Dongming Qiao*, Guanqi He*, Zhenyu Wang, Yitang Li, Shaoting Zhu, and Hang Zhao.
*arXiv:2606.10340*, 2026. Paper | Project | Code
RSS 2026 workshops: ExWBC (**Oral**); RoboData (**Spotlight**).

**Hiking in the Wild: A Scalable Perceptive Parkour Framework for Humanoids**
Shaoting Zhu*, Ziwen Zhuang*, Mengjie Zhao, **Kun-Ying Lee**, and Hang Zhao.
**Conference on Robot Learning (CoRL), 2026** (accepted). Paper | Project | Code

**The Indefinite Integral of the *n*th Power of Inverse Trigonometric Functions**
**Kun-Ying Lee**.
*Mathmedia*, Academia Sinica, 49(2):91–101, 2025. Article

* Equal contribution.

## Research Experience

**MARS Lab, Tsinghua University** 2025–present
*Research Assistant | Advisor: Prof. Hang Zhao*

- **OMG-Posttrain.** Developing reinforcement learning methods for post-training OMG, focusing on closed-loop execution of highly dynamic motions with asynchronous motion generation and control.
- **OMG.** Developed a scalable, omni-modal humanoid foundation model that maps language, audio, and human reference motions to executable whole-body motions on Unitree G1.
- **Hiking in the Wild.** Validated sim-to-sim policy transfer and assisted with real-robot deployment of a depth-conditioned humanoid parkour policy.

## Engineering Experience

**TH-MOS Robot Soccer Team, Tsinghua University** Sept. 2025–present
*Role: Gait Control Group Lead*

- Lead soccer-skill development and train policies for omnidirectional velocity tracking, passing, shooting, goalkeeping, and fall recovery.
- Built a **MuJoCo**-based RoboCup simulation environment integrating decision-making, locomotion control, and perception for rapid policy evaluation and iteration.

## Selected Honors & Awards

**5th Place**, RoboCup 2026 Incheon (TH-MOS team) 2026
**2nd Place**, RoboCup Asia-Pacific Tianjin Invitational Tournament (550W team) 2026
**3rd Place**, RoboCup China Open (TH-MOS team) 2026
**Leading Talent (Lingjun) Award**, Tsinghua University 2025